Glossary

Agency
The capacity of individuals to act or exert influence on their surroundings. In historical and social contexts, agency is often discussed in terms of an individual's relationship to power structures or social dynamics that enable or constrain one's ability to act. Agency can be analyzed at the level of grammar as it is represented through sentence structure—specifically, who is positioned as the subject (the one performing the action) and who is the object (the one receiving the action). This linguistic framing can shape perceptions of power, responsibility, and control in discourse.

Archives
Collections of historical records, documents, or data that are preserved for reasons such as cultural heritage or research. Archives can be physical (e.g., government records, letters, manuscripts) or digital (e.g., databases, repositories). Notable digital archives for historical research include HathiTrust, Internet Archive, Project Gutenberg, and Chronicling America. These repositories provide access to historical texts, newspapers, and various multimedia.

Argument, Function
In programming, an argument refers to a value or variable that is passed to a function. A function is a reusable block of code designed to perform a specific task. Analysts typically write functions to keep from rewriting the same code repetitively, as the function name can be used instead of rewriting code line-by-line. A function typically consists of three key parts: a name, a list of input arguments (also called parameters), and a body—the set of code instructions or operations that define what the function does. When the function is called or "invoked," the arguments are passed in, and the body code is executed using those values. The function may return a result, often referred to as the output.

Britain / British Empire
In the context of the nineteenth century, Britain refers to the United Kingdom, particularly England, Scotland, Wales, and Ireland. The British Empire (c. 16th–20th century) was a global colonial and imperial entity, exerting political, economic, and military control over vast territories across Africa, Asia, the Americas, and Oceania. Parliamentary debates during this period—including discussions of imperial policy and colonized regions—are recorded in Hansard, the official transcript of proceedings in the British Parliament.

Cleaning
The process of preparing and refining data for analysis by identifying and removing inconsistencies, errors, or irrelevant data that might detract from the analysis. While cleaning may be essential in most computational research, it is not a neutral process. Decisions about what to remove or what to retain involves the analyst's judgment and can significantly shape the outcomes of analysis. Cleaning can improve the quality of data for certain types of analyses, but it also risks introducing distortions, biases, or reinforcing assumptions that may obscure the complexity of the original source.

Code
A structured set of instructions written in a programming language that can be used for many purposes including data analysis and automation of repetitive tasks. Code serves as a way for humans to communicate with computers, translating data processing steps into operations a machine can execute. The structure of code typically includes variables, functions, conditionals, and loops. Well-written code is often reusable and accompanied by comments to improve readability and collaboration.

Coding Environment
A software platform or integrated development environment (IDE) where analysts write, test, and run code. Coding environments often provide helpful functionality for writing code such as syntax highlighting and

debugging. They may be language-specific (e.g., RStudio for R) or support multiple languages. Just a few more examples of coding environments include: Visual Studio Code, Jupyter Notebook, and Google Colab. Google Colab also supports co-work in that it allows multiple people to access a single notebook at a given time.

Comments (in Code)
Annotations written within code to describe specific sections. Though they do not affect how the program runs, comments are useful for documenting reasoning, outlining structure, and making notes to oneself. They can capture the analyst's assumptions or interpretive choices, providing a layer of meaning parallel to the programming logic.

Concordance
A structured list of all occurrences of a specific word or phrase within a text or corpus, typically displayed with a snippet of surrounding context (known as "key word in context," or KWIC). Concordances allow researchers to examine patterns of language usage and variation across different sources or time periods. They are foundation tools in both linguistic, literary, and historical analysis, helping researchers contextualize words and trace shifts in meaning or the linguistic framing in which words are presented. In historical text analysis, concordances can offer an entry point into the interpretive labor of distant reading, or an intermediary transition point between a distant reading and can also act as bridges to more focused close readings of full documents.

Corpus vs. Dataset
A corpus is a collection of texts assembled for analysis, often used in linguistic, literary, or historical research. A corpus is usually selected from a specific genre, time period, or domain. For example, the NovelTM corpus (Underwood et al.) includes thousands of English-language novels for computational literary analysis, while the Hansard Corpus contains transcriptions of British parliamentary debates from the 19th and 20th centuries for political and historical research. Digital corpora used in computational research often undergo natural language processing techniques before they are analyzed. This may include tokenization, part-of-speech tagging, syntactic parsing, or the addition of structured metadata such as author, date, and publication context. A dataset, by contrast, is a broader term that refers to any organized collection of data—textual, numerical, categorical, or otherwise—structured in a way that supports computational analysis. While all corpora are datasets, not all datasets are corpora. The distinction stems from differing priorities: corpora emphasize language and interpretive context, whereas datasets are defined more generally by format and function, and may prioritize measurement or modeling over textual interpretation.

Corpus Linguistics
The study of language using large, structured collections of texts (corpora) that are analyzed with computational and statistical methods to identify patterns across many texts rather than relying on close readings of isolated examples. Methods from corpus linguistics enable researchers to identify patterns such as word frequency, syntactic constructions, and semantic shifts over time. For example, scholars like Marc Alexander and Jessie Demmen have used the Hansard Corpus—over 1.6 billion words of British parliamentary debate stretching back to 1803—to examine how language reflects shifting attitudes toward national identity, race, and empire. Their work demonstrates how corpus linguistics can characterize political discourse across centuries.[1] Corpora are typically annotated with metadata (e.g., date, genre, speaker) and may undergo natural language processing to include linguistic features such as part-of-speech or syntactic tags. In historical and cultural research, corpus linguistics is especially valuable for tracing diachronic shifts in meaning. For example, researchers might use corpora to examine how the lexical field surrounding terms like liberty, race, or gender changes across decades or centuries.

Critical Evaluation (of Code)
A systematic assessment of code's suitability in relation to the interpretive and methodological aims behind a given historical question. This process includes analyzing the underlying steps of the code, as well as validating its outputs against expected or theoretically grounded results. In history research, such evaluation extends beyond functionality to include questions of epistemological rigor including whether the computational approach is appropriate for the research objectives, supports defensible interpretations, and engages responsibly with the available evidence.

Critical Search
Using critical thinking to engage with data by putting them into conversation with primary and secondary sources and theory. To critically search is to engage in an iterative process of asking questions with data, rather than to execute a single algorithmic procedure and to treat the output as an intervention. Critical search may be conceived of as movement between "seeding" (using the known literature to propose a series of questions that a dataset can help to answer and how), "broad winnowing" (successive passes through the data, which iteratively narrow the body of work to be consulted), and "guided reading" (where the analyst turns to carefully read primary sources, guided by findings in the dataset).

Critical Thinking (about History)
Guides almost every stage of explaining what happened in the past and analyzing change over time. The analyst must ask of each primary and secondary source: what is the bias that this source carried? Questions such as these afford the opportunity to ask who participated in, experienced, or caused different episodes of change over time, and whether the event was experienced the same way by each of them. Each event, in turn, is subject to the same questioning process: when did it begin? When did it end? Do we understand the event differently if we look over a longer or shorter period of time? Critical thinking about the past is impossible without an appreciation of the primary textual sources that lead to interpretation as well as an understanding of how other scholars have seen the event in the past, and these pressures put real constraints on any form of digital analysis divorced from the interpretive concerns of historians.

Critical Theory
An interdisciplinary and evolving intellectual tradition that includes a wide range of perspectives for analyzing the construction of knowledge and the role of culture in shaping social life and cultural artifacts, like text. While often associated with the critique of power—examining how systems of authority and inequality are embedded in language and institutions—critical theory also engages with questions of aesthetics, ethics, and identity. Critical theory includes diverse schools of thought such as Marxist theory, feminist theory, postcolonial theory, critical race theory, queer theory, and deconstruction. Each of these offer different approaches for interpreting texts and understanding the social world. When critical theory is applied to histroical corpora such as the Hansard debates it can support researchers in exploring how parliamentary language helped constitute political realities—reinforcing imperial hierarchies, gender norms, or liberal ideals of citizenship. A Marxist reading might trace how economic interests and class ideologies shaped debates about labor or land; a postcolonial perspective might focus on how the discourse of civilization was mobilized to justify colonial violence; while a feminist analysis could reveal how women were rendered powerless in legislative speech.

Cultural Analytics
Approaches to the study of culture through computational methods, data analysis, and visualization tools. Cultural analytics bridges the concerns of the humanities, cultural studies, and data science, and frequently requires inspecting data about cultural artifacts – such as text, images, music, film, or social media – at scale.

Data
Provides the evidentiary basis for analysis. Data consist of observations, measurements, or representations that can be systematically analyzed. They may be quantitative, such as numerical values that can be counted or measured, or qualitative, such as interview transcripts, ethnographic notes, texts, images, or other forms of descriptive and contextual information. The term "data" also invites critical scrutiny—not only of how information is collected and encoded, but of the epistemological and ethical assumptions that underlie its use. This is because data collection is not a neutral task–it is shapped by human choices on what is collected, how it is categorized, and what is excluded. In New Digital Worlds, for example, Roopika Risam argues that data must be understood as a product of cultural and historical processes. As Risam notes, the design of data curation reflects power relations and value systems, and thus requires ongoing critical reflection—particularly in projects that engage marginalized histories and cultural memory.[2]

Historical Data
Information drawn from records, documents, artifacts, and other sources that provide insight into the past. Historical data is foundational to fields such as history, digital humanities, archival studies, and cultural analysis—but its relevance extends well beyond academia. Industries such as finance, insurance, energy, and

urban planning rely on historical datasets to track trends over time. For example, shipping logs and climate records inform supply chain logistics; census data underpins market research and demographic forecasting; and historical patent filings shape our understanding of technology. Historical data may be structured, like tax records or parliamentary transcripts, or unstructured, such as oral histories or marginalia. Researchers working with historical data face persistent challenges, including gaps in the record, biases in what was preserved, and distortions introduced through digitization. Working with historical data demands an awareness of the ethical and epistemological implications of digitization, particularly as historical silences and exclusions of people are reproduced in digital corpora.

Data Science vs. Computer Science vs. Information Science
* Data Science is an interdisciplinary field focused on creating knowledge from data. It uses computational modeling techniques such as statistical analysis or machine learning to identify patterns. Data science can be both methodological and applied, emphasizing not only the development of algorithms but also their interpretation. Its scope spans a variety of domains—including public health, climate science, economics, and the humanities—where complex, often unstructured data must be transformed into structured forms for analysis. Data science draws from computer science for its infrastructure and tools, and from information science for concerns around data provenance, quality, and stewardship. * Computer Science is a foundational discipline concerned with the principles and technologies that enable computation. Computer science includes both theoretical and practical dimensions: the former includes areas such as automata theory, formal languages, and computational complexity, while the latter includes software engineering or and human-computer interaction. Computer science tends to be less concerned with the content of data and more with the logic, structure, and efficiency of computational processes themselves. * Information Science examines the life cycle of information—its creation, organization, dissemination, retrieval, and use—within human and technological systems. Information science is inherently socio-technical, concerned not only with information infrastructures like databases and digital archives but also with the social dimensions of information access and management. Information science engages with questions of data classification, metadata standards, privacy, and information equity. It has deep historical ties to library and archival studies but has evolved to include the study of digital platforms, user experience, and information policy.

data frame
A two-dimensional, tabular data structure widely used in digital historical research for organizing and analyzing structured data. Commonly implemented in the R language, a data frame arranges information into rows and columns, where each row might correspond to a discrete historical observation—such as a census record, parliamentary speech, notarial act, or newspaper article—and each column represents a specific variable, such as date, location, or author. This format supports systematic analysis of large-scale historical datasets including comparisons across time periods or sources. Researchers can apply filters to data frames or create visualizations.

Data Type
The classification of data based on its characterizations and the kinds of programmatic operations that can be performed on it. Common data types include numerical data, such as integers and floating-point numbers, which are used for calculations and measurements. Categorical data represents discrete groups or labels and can be either nominal (without order, like country names) or ordinal (with a meaningful order, like rankings). Boolean data captures binary values such as true or false, often used in logical operations. Text data, or strings, consists of characters and is used to represent words or sentences. Date and time data captures temporal information, including timestamps and durations, allowing for chronological analysis and time-based comparisons.

Digital Humanities
An interdisciplinary domain that integrates computational methodologies with humanistic inquiry as part of an interpretive process. Digital humanities research may employ techniques such as text mining, data visualization, spatial analysis, and network analysis, enabling scholars to explore cultural, historical, or literary questions at scale while also attending to humanities interpretation. The field has its roots in the mid-20th century tradition of "humanities computing," which stems from Father Roberto Busa's collaboration with IBM to produce a digital concordance of Thomas Aquinas's works.[3] In the early 21st century, digital humanities expanded significantly, shaped by institutions such as the Stanford Literary Lab, co-founded by

Franco Moretti. Moretti's concept of "distant reading" advocated for the use of quantitative methods to analyze large corpora of literature. This approach, and others like it, challenged traditional methodologies in literary studies by treating literature as a dataset to be modeled.[4] Today, digital humanities encompasses a wide range of practices—from building digital archives to designing interactive platforms and a space for reflecting on the consequences of computational research in the humanities.

Distinctiveness (e.g., TF-IDF)
A statistical principle used to evaluate the significance of a word in a specific document relative to a larger collection of texts, or corpus. Words receive higher scores when they appear frequently in a specific document but are relatively uncommon across the corpus, helping distinguish unique terms from common words. TF-IDF was originally developed in the 1970s by information retrieval researcher Karen Spärck Jones as part of her pioneering work in information retrieval—that is, the science of designing systems (like search engines) that help users find relevant documents in large collections of text.[5] One of the central challenges in information retrieval was determining which documents were most relevant to a user's query. Simply counting how often a word appeared in a document was not enough because many words occur frequently across large collections of texts. Karen Spärck Jones addressed this problem with inverse document frequency (IDF), which assigns less weight to words that appear in many documents and more weight to words that are relatively rare. Combined with term frequency (TF), TF-IDF highlights words that are especially important to a particular document. In computational historical analysis, it can help identify terms that are unusually prominent in specific speeches, debates, or texts relative to broader patterns of language use.

Distance (e.g., JSD)
A mathematical measure for quantifying the difference or dissimilarity between two sets of language data. Divergence metrics are widely used in computational text analysis to track how word usage shifts across corpora, time periods, or social contexts. One prominent example is Jensen–Shannon Divergence (JSD), a method that measures how two distributions—such as word frequencies—differ. In digital historical research, JSD has proven especially meaningful for studying long-term changes in discourse. For example, Sara Klingenstein, Tim Hitchcock, and Simon DeDeo used JSD in their study "The Civilizing Process in London's Old Bailey" to examine how courtroom language changed over more than a century of trial transcripts. By comparing distributions of words used in earlier versus later trials, they demonstrated a gradual shift from violent and physical descriptions of crime to more abstract and moralizing language—evidence of what they interpret as a "civilizing" transformation in legal discourse.[6]

Digitization
The process of converting physical records—such as manuscripts, printed books, or maps—into digital formats that can be stored, accessed, and analyzed with a computer. Digitization ranges from basic scanning, which generates static image files, to techniques such as optical character recognition (OCR), which transforms printed or handwritten text into machine-readable data. Advances in large vision models hint at greater transcription accuracy, especially for handwritten and multilingual documents as large vision models are able to interpret more complex layouts or faded and irregular handwriting with greater precision than traditional OCR tools. For instance, projects such as Transkribus have begun to incorporate deep learning for the transcription of early modern handwriting.

Embeddings
Embeddings are a way of converting words or documents into numbers that represent their meaning based on how they are used in language. Each word is assigned a location in a mathematical space with many dimensions. In this space, words that are used in similar contexts are placed closer together, while words that are used in different contexts are placed farther apart. The distance between words therefore provides a quantitative measure of their similarity in meaning and usage. Researchers can analyze the resulting model and identify relationships between words based on how they are used in context within a text or corpus. When analyzing change over time, word embeddings can reveal how associations between words shift across different historical periods. For example, they might show that in the United States Congressional Record, the word "man" became increasingly associated with the term "gay" in later decades—indicating a change in how certain concepts were linked in political discourse.[7]

Event
A discrete unit within a dataset that signifies a notable action or occurrence. In historical research, an event

typically refers to a documented moment—such as a political speech, the passage of legislation, a protest, a battle, or the publication of an influential text—that signals a transformation in social or cultural dynamics. In computational analysis, events may be identified from either structured data or unstructured sources. Researchers who analyze events can identify historical patterns and situate individual occurrences within larger historical processes.

Field (in Academic Study)
An organized area of scholarly research defined by a common set of questions, methods, theories, and subject matter. Fields can range from broad disciplines like history or computer science to more specialized areas like digital humanities or forensic linguistics. Fields provide a framework for how research is conducted, such as what tools and approaches are prioritized. They also influence how scholarly work is received and evaluated by an audience. Fields also structure academic communities by forming networks of journals and conferences. Over time, fields may shift or expand as new topics gain attention, interdisciplinary work emerges, or social and technological changes influence research priorities.

Field (in a Dataset)
A column or attribute in a structured dataset that represents a specific category of information. For example, in a dataset of historical speeches, fields might include "Date," "Speaker," "Location," and "Debate Title" Each field contains a particular type of data—such as text, numbers, or dates—and helps organize the dataset in a consistent, analyzable format. Fields are used for filtering and sorting data for quantitative or qualitative analysis.

Filter (in a Dataset)
A process used to refine or subset a dataset by applying specific conditions to one or more variables in a dataset. Filtering narrows down large datasets to focus on relevant sections. By using filters, researchers can extract only the records that meet certain criteria, such as selecting speeches delivered by a particular politician, documents from a specific decade, or entries that mention a key term.

GitHub
A web-based platform for version control and collaborative programming, built on the Git system, which tracks changes in code and text files over time. Many R packages relevant to historical data analysis—such as hansardr (for British parliamentary debates), usdoj (for U.S. Department of Justice data), and noaa (for climate and weather datasets)—are hosted on GitHub, making them openly accessible. Along with providing access to data, code, and documentation. In this way, GitHub advances the principles of open scholarship and aligns with FAIR data guidelines (Findable, Accessible, Interoperable, and Reusable), helping support digital research.[8]

Grammar
The system of rules that governs the structure and use of language, encompassing elements such as syntax (sentence structure), morphology (word formation), and semantics (meaning). In linguistic and historical analysis, analyzing grammar can support interpretations about how power dynamics and ideological positions are embedded in historical texts. Subject-object relationships, verb tense, and voice (e.g., active vs. passive) shape how actions are attributed and events are framed.

Hansard
The official transcript of parliamentary debates from Great Britain, Hansard documents the speeches and decisions made in the UK Parliament and serves as a key resource for historical and political research. Multiple versions exist for different scholarly purposes. The Historic Hansard (1803–2005) is publicly available through the UK Parliament, while curated versions like those by Egger and Spirling or the hansardr R package offer cleaned and structured data for computational analysis.[9] Proprietary versions, such as the version of Hansard hosted by ProQuest, provide additional metadata and search functionality but often require institutional access.

hansardr
An R data package that provides easy access to 19th-century UK Hansard parliamentary debates within R. Available through GitHub, hansardr includes analysis-ready versions of the debates organized by decade, allowing researchers to work with manageable partitions of the corpus rather than loading the entire collection

into memory. This structure makes it possible to analyze large-scale parliamentary records on a personal computer.

Historiography
Historiography is the study of how history has been written about over time. Historians rarely agree about which events are important, who participated in those events, what caused them or what their most impactful consequences were. Reviewing the tensions between different perspectives is the work of historiography. In the professional discipline of history, reviewing historiographical debates is often understood as a necessary preliminary step before studying new historical material, primary sources, or data – because only after considering a perspective on how other scholars have interpreted the past, agreed or disagreed in the past, can a scholar be confident in a new interpretation.

History
History is the study of the past, especially through the examination of change over time as marked by "events," where culture, society, ideas, politics, or economics shifted in a way noticeable to participants. History also encompasses the study of periods of time, which are stretches of time punctuated by events. Periods and events may both be long or short. Widespread consensus about the major episodes in collective history has long been understood as a necessary precondition for a society to accept a shared form of governance, especially democracy. At the same time, dissent about history is typically considered productive of integrating new points of view into a perspective on the past, so long as the dissent in history is based on facts – that is, forms of evidence that survive from the past, whether in the form of documents, landscapes, data, or folklore – whose origin in the past is recognized as valid by most careful observers, whether or not those observers share other interpretations of the facts' consequence.

Iteration
A repeated process of refinement and revision. Iterative research processes are central to both computational research and historical analysis. In data science, iteration often involves adjusting algorithms, code, models, or visualizations to see how results change. In historical research, iteration may take the form of moving between different layers of evidence—starting with a visualization or summary of data, then returning to the original dataset, consulting relevant secondary literature, and ultimately reexamining primary sources. In the research demonstrated by this book, analysts might play a hybrid role by continually revisiting and refining their understanding of the historical documents and the method through which they arrived and their understanding.

Joins
A data processing operation that combines data from multiple tables based on a shared column. Joins are a key part of data manipulation in R, particularly when using packages like `dplyr` (from Hadley Wickham's `tidyverse`). Common join types in R include **inner_join()**, which returns only the rows with matching keys in both datasets, and **left_join()**, which preserves all rows from the first dataset while merging in any matching data from the second.

Keywords
Significant words or phrases used to index, search, or analyze texts. Keywords can guide digital historical research. Their notion stems from cultural theorist Raymond Williams who famously explored the layered and contested meanings of socially significant words in his book **Keywords: A Vocabulary of Culture and Society**, arguing that certain terms serve as windows into broader ideological, cultural, and historical shifts.[10] Building on this foundation, keywords are used by historians as conceptual anchors to trace changes in discourse, examine the evolution of political or cultural values, and understand how language reflects and shapes social life. In computational research, keywords can be manually assigned—such as metadata tags in digital archives—or automatically extracted using methods like TF-IDF (Term Frequency-Inverse Document Frequency), which identifies terms that are especially distinctive within a given document or corpus.

KWIC (Key Word in Context)
A method in text analysis that displays a selected keyword alongside a fixed number of words before and after it, providing a window into the term's immediate context like a concordance. KWIC concordances are especially useful for examining how words are used in a document by revealing patterns in meaning, tone, and grammatical usage across different texts. KWIC is widely used in linguistic, historical, and digital

humanities research to trace shifts in discourse over time. In R, the `quanteda` package offers a built-in `kwic()` function that allows researchers to generate KWIC displays quickly from large corpora.[11]

LLM (Large Language Model)
A form of artificial intelligence (AI) trained on large amounts of text to generate, interpret, and analyze language. Proprietary examples include OpenAI's GPT-4 and GPT-3.5, Google's Bard, and Anthropic's Claude. Open source examples include Meta's Llama 3 or Mixtral. In historical research, LLMs can be used for code generation, metadata extraction, and exploratory analysis of large corpora. Despite their impressive ability to respond to human input, LLMs do not truly reason. Their outputs are probabilistic, based on predicting likely word sequences rather than understanding meaning or context. As a result, while LLMs can be valuable tools, their outputs must be critically evaluated, as they may contain inaccuracies such as "hallucinations" (responses that are semantically plausable but factually incorrect), reflect biases, or omit the complexities of historical evidence.

Longue Durée, Big History, Moyenne Durée, Microhistory
Longue-durée history is history of long spans of time – as short as a few decades or as long as multiple centuries, although typically history longer than a few millenia is classified as "big history," because its analysts must have recourse to sources from evolution and neuroscience, rather than the documentary set of sources upon which most modern historians rely. In theory, the analysis of datasets can help analysts to expand their account of history over decades to a century or more, although in reality, most text-based datasets linked to institutions (such as Hansard) are limited to a few decades or at most two or three centuries. Approaches that aggregate information across time and space at scale are contrasted with microhistory, an approach to social history that developed in Italy and France in the 1960s, and which emphasizes the intensive reading of local archives in order to develop a rich account of the past. Originally, local microhistories made room for longue-duree analysis, although social historians of the 1980s, 1990s, and 2000s tended to gravitate to the examination of local history over timescales as short as a year or several years.

Metadata
Descriptive information about data that provides context and structure and transforms documents into searchable, analyzable datasets. In short, metadata is data about data. In the context of Hansard, metadata includes details such as the name of the speaker, the date of the debate, the debate title. This metadata allows researchers to filter speeches by speaker or trace debates across specific time periods. For historians, metadata in Hansard supports both targeted searches—such as locating all speeches by a particular MP—and large-scale computational analysis, supporting comparisons across decades.

N-grams (Unigram, Bigran, Trigram)
Sequences of n words or characters often used in computational text analysis to examine patterns, co-occurrence, and linguistic structure. N-grams are derived through the process of "tokenization" or breaking text into meaningful units, which can be individual words, phrases, or characters.

- Unigram (1-gram): A single word (e.g., history).
- Bigram (2-gram): A two-word sequence (e.g., digital humanities).
- Trigram (3-gram): A three-word sequence (e.g., machine learning model).

Normalization
Normalization is a data processing technique used to standardize data to ensure consistency and comparability across a dataset. In computational text analysis normalization typically involves transforming text to a uniform format by applying steps such as lowercasing all words, removing diacritics (e.g., converting *résumé* to *resume* or vice-versa), and using lemmatization or stemming to reduce words to their base or root form (e.g., if lemmatizing, run, running, and ran are each transformed into run). The goal may be to minimize variation of word forms that could otherwise impact frequency counts or other forms of computational pattern recognition. In historical research, normalization allows scholars to compare terms across different time periods, spelling conventions, or textual sources, making it a common early step for an analysis.

Operating System (OS)
The software that manages a computer's hardware and software. Common operating systems include Windows, macOS, and Linux. Each OS offers different capabilities and interfaces. In historical analysis or data

science, the choice of operating system can impact software installation or data processing. For example, certain R or Python packages may require Unix-based systems (like macOS and Linux) to install correctly, and command-line tools often function differently across OS platforms. Understanding OS compatibility is therefore important for managing software dependencies.

Parliament
A legislative body responsible for debating, shaping, and enacting laws within a government. It serves as a central institution in many democratic systems, often comprising elected representatives who engage in public debate and decision-making. Parliamentary records, such as the British Hansard, are primary sources for historical and political research. Hansard provides detailed accounts of legislative proceedings, including speeches, debates, and policy discussions, offering insights into how laws were formed and how political ideologies evolved to influence public opinion and governance over time.

Pedagogy
The methods, design, and underlying philosophy that guide how knowledge is structured and delivered to learners. Pedagogical approaches are often aligned with the values and practices of the discipline itself.

Periodization
The practice of dividing history into distinct time periods based on shared social, cultural, political, or economic characteristics. The divisions that result from periodization—-such as the Renaissance, Enlightenment, or Industrial Revolution-—help historians organize historical developments and make sense of change over time. However, periodization is not a neutral process; it often reflects the priorities and interpretive frameworks of a historiographical tradition. Therefore, while useful for structuring narratives and comparative analysis, period labels can obscure continuities or impose artifical boundaries that marginalize human experiences. As such, periodization is both a tool for analysis and a subject of critique.

Plots
Visual representations of data that make it possible to view patterns, trends, and relationships that might be difficult to detect in raw numerical or textual data. They are useful tools in both exploratory analysis and the communication of results as they allow researchers to summarize complex information in a visual fashion. In historical research, plots can be used to track change over time—for example, a line plot might show how the frequency of a political term changes across decades, while a bar chart might compare the number of speeches given by different political parties in a given period. Categorical data, which refers to variables that represent distinct groups or labels (such as speaker names, political affiliations, or geographic regions), is often visualized through bar charts or grouped scatter plots. Other plots, like histograms, can illustrate the distribution of values, such as word counts or sentence lengths in a corpus.

Prediction
A computational process that estimates future values, classifications, or outcomes based on patterns in existing data. In machine learning, algorithms are trained to identify trends. Methods of prediction include statistical forecasting, such as projecting economic indicators over time; text classification, such as identifying the likely sentiment or political leaning of a speech; and topic modeling, where prediction refers to estimating the probability that certain latent themes are present in a document based on word co-occurrence patterns. Predictive methods must be used with caution in historical contexts, as language is shaped by cultural and temporal factors that do not necessarily follow consistent or repeatable patterns. Words in a historical corpus do not "predict" future usage in a statistical sense, and applying predictive models without context risks oversimplifying complex historical phenomena. As such, prediction in historical research is best understood as a tool for exploring patterns and generating interpretive questions, rather than as a method for drawing definitive conclusions about the future based on the words used in the past.

Primary Sources
Original documents or artifacts that offer firsthand accounts of historical events, created by individuals who experienced or experienced or observed them directly. Primary sources include manuscripts, letters, diaries, newspapers, government records, oral histories, photographs, and material culture. In the context of political and institutional history, **Hansard**, the official transcript of British parliamentary debates, serves as a primary source that captures the language of legislators. Unlike secondary sources, which analyze or interpret historical evidence, primary sources provide direct access to the voices and contexts of the past.

Programming Language
A formal system of symbols used to instruct computers to perform specific tasks, ranging from data processing to app development. Different programming languages are better suited to different tasks because they have different features, ways of organizing code or processing data, and communities that create tools, libraries, and other support resources around them. In data science and history, languages like Python and R are widely used for tasks such as text analysis, data cleaning, and visualization. SQL may be used for querying and managing structured data in databases, making it useful for working with archival metadata or large corpora. For building web-based projects or interactive visualizations, languages such as JavaScript, HTML, and CSS are commonly used. The choice of programming language shapes both the capabilities of a project and the ways researchers engage with their data.

R
A programming language and open-source software environment originally developed in the early 1990s by statisticians Ross Ihaka and Robert Gentleman for statistical computing and data analysis. R has since become a widely used language in history and data science for analyzing large textual corpora, modeling patterns in historical data, and producing reproducible research. R's extensive ecosystem of packages—especially those in the tidyverse—allows researchers to clean, structure, process, and visualize data with syntactic and stylistic consistency. Tools like quanteda, tidytext, and tm support text mining, keyword analysis, topic modeling, and other forms of computational textual interpretation. Its community and emphasis on transparency and openness make R especially valuable for open science projects that combine humanistic inquiry with data science methods.

RStudio
An integrated development environment (IDE) for R that provides a comprehensive interface for writing, debugging, and running R code. RStudio uses multiple panels to organize the development environment for analysts. The Source panel allows users to write and manage R scripts, while the Console panel is where code is executed interactively. The Environment/History panel tracks the objects in the current session and maintains a history of commands. The `Files/Plots/Packages/Help` panel offers access to file directories, visualizations, installed packages, and R documentation. The panels in RStudio work together to provide a cohesive environment to support data analysis. Additionally, RStudio supports interactive data visualization for exploring data through charts and plots.

Sampling (vs. Subsetting or Filtering)

- Sampling: The process of selecting an often representative subset of data from a larger dataset, often used in analysis to infer trends from a smaller portion of data. Common methods include random sampling (selecting data points at random) and stratified sampling (dividing the population into subgroups and sampling from each to ensure representation).
- Subsetting: The process of selecting a portion of a dataset based on specific criteria, but without necessarily being representative of the whole. For example, a researcher might subset data to focus on all records from a particular year or region.
- Filtering: A specific form of subsetting where only rows meeting a defined condition are retained. For example, filtering a dataset to include only texts written by a particular author or only data points with a value above a certain threshold.

Secondary Sources
In historical research, secondary sources are scholarly works that interpret or analyze primary evidence to construct arguments about the past. Rather than offering firsthand accounts, secondary sources draw upon archival materials, official records, newspapers, and other primary documents to provide context, critique, and narrative. Historians rely on secondary sources to engage with existing interpretations and to situate their work within broader scholarly debates. Such sources are commonly found in academic monographs published by university presses, articles in peer-reviewed journals indexed in databases like JSTOR, Project MUSE, or ProQuest Historical Newspapers, and dissertations cataloged through platforms such as the MLA International Bibliography or WorldCat.

Snippets
Short extracts or sections of text, code, or data that are used for quick reference or demonstration. Snippets allow for quick exploration of concepts or examples without needing to engage with an entire dataset or program. In text analysis, snippets may provide contextual examples of how keywords or phrases appear within larger bodies of text, helping researchers understand their usage and significance. In programming, code snippets illustrate small, functional segments of code, often shared to demonstrate specific techniques or solve particular problems.

Social Sciences
The study of human behavior, societies, and institutions using empirical research methods. Key disciplines within the social sciences include sociology, anthropology, political science, economics, psychology, and communication studies. These fields use both qualitative and quantitative methods to analyze social phenomena. The findings from these fields may inform social policy and theory. History is sometimes viewed as a social science because it uses similar empirical methods to study societal behavior and institutions. When historians analyze past social, political, or economic trends using data science approaches or statistical tools, history aligns with the social sciences by seeking to understand patterns and causes of historical events.

Software Package / Library
A collection of prewritten code designed to perform specific actions. Packages may be designed by professionals or community members to extend the capabilities of a programming language without needing to write new code from scratch. A library is often a specific type of package that provides reusable functions, typically organized around a particular domain or task, such as natural language processing or data visualization. Examples of packages include `tidyverse` (R), which is used for data manipulation and visualization, and `spaCy` (Python), a library focused on natural language processing. `ggplot2` (R) is another example, a package that provides functionality for data visualization, particularly in the `tidyverse`.[12]

Stop Words
Stop words are common words like “the,” “and,” “of,” or “is” that are often removed in text analysis because they carry little meaning on their own and appear frequently across datasets Stop words are often removed because they can obscure the more meaningful parts of the text. For example, without removing stop words, a word frequency visualization might be dominated by common words like “the,” making it difficult to highlight the more relevant or interesting terms. However, the decision to remove stop words is not without its challenges. What qualifies as a stop word can be subjective, depending on the context and the goals of the analysis. Removing stop words can sometimes distort the data by omitting important context that might affect the text’s meaning. For instance, many popular natural language processing tools, such as Stanford’s NLP toolkit, remove gender pronouns (e.g., “he,” “she”) by default as part of their stop word lists. While this can be useful for certain types of analysis, it may be misleading when analyzing texts where gendered language is critical to understanding the content. Therefore, researchers must be critical and selective in deciding which stop words to remove, as this decision can significantly influence the results and interpretations of the analysis.

Subsetting
The process of extracting a portion of a dataset based on specified conditions. Unlike sampling, which often is used to create a representative selection, subsetting focuses on isolating specific rows, columns, or values that meet certain criteria for further analysis. This method is useful when researchers want to examine particular partitions of data, such as records from a specific time period, a certain category, or values within a defined range.

Syntax (in Code)
The set of rules that define the structure and format of a programming language. It governs how commands, functions, and expressions must be written in order for the code to execute properly. For example, in R, the correct syntax would be, `my_data <- read.csv("data.csv")`, while the following would be incorrect due to improper syntax, `my data = read csv("data.csv")`. “my data” has a gap between text, which would result in an error. Instead of using the `<-` assignment operator we used `=`, which technically works but is stylistically incorrect for this context. Instead `=` is typically used in function calls. Incorrect syntax can lead to errors, preventing the code from running or producing unintended results. Unconventional syntax can lead to miscommunications between groups working on shared code.

Syntax (in Text)
The arrangement of words and phrases to form grammatically correct sentences in a human language. It dictates how different elements of a sentence—such as subjects, verbs, and objects—are structured to convey meaning clearly and coherently. Syntax plays a crucial role in natural language processing because it influences how meaning is represented within text. In computational text analysis, software like `spacy` are commonly used to analyze sentence structure and perform part-of-speech tagging, which identifies the syntactic roles of words (e.g., noun, verb, adjective). In **Text Mining for Historical Analysis** syntactic analysis is used to understand the relationships between parts-of-speech in historical texts and demonstrate how analyzing parts-of-speech can provide insight into gender and power dynamics.

Text Mining vs. Generative AI
Text mining involves extracting patterns or structured data from (often large) text corpora using computational techniques such as word counts, TF-IDF, named entity recognition, and topic modeling. Text mining is primarily used to analyze existing text data to engage with language features within the data. On the other hand, Generative AI refers to AI systems that generate new text, images, or data based on patterns learned from training data. Understanding the differences between typical text mining approaches and generative AI is important, as generative AI can return incorrect assumptions about history using a confident sounding tone as the generated responses are not bound by the original data like in text mining.

Tidy Data
A structured data format designed to simplify analysis and visualization. In tidy data, each column represents a variable, which is any characteristic or feature that can vary across each row. For example, in historical analysis, variables could include the date of a speech, the speaker's name, or the location of an event. Each row represents an observation, which is a single instance or record within the dataset—such as a specific speech given by a politician or a historical event on a given day. Each cell contains a value, which is the actual data point or measurement for a given variable and observation, such as the length of the speech or the word frequency of key terms in the speech. This structure makes data easier to process, read, and visualize, and it is particularly useful when using tools like the tidyverse in R, which is designed to work with tidy data formats.

Tidyverse
A collection of R packages developed primarily by Hadley Wickham that are designed to support data science through a unified syntax for data processing. The Tidyverse has significantly influenced contemporary data science practice by promoting a philosophy in which data structure is treated as foundational. Central to its design is the concept of "tidy data," a set of principles articulated in Wickham's 2014 paper "Tidy Data."[13] Tidy principles assert that each variable should form its own column, each observation its own row, and each type of observational unit its own table. This structure facilitates clarity and interoperability across packages. Core Tidyverse tools—including (but not limited to) dplyr for data manipulation, tidyr for reshaping data, and ggplot2 for visualization—operate most effectively when data conforms to this tidy format. Together, the packages making up the `tidyverse` have reshaped how researchers conceptualize and engage with data.

Vector
A data structure in programming and data analysis, representing an ordered list of values. Vectors can be of different types, such as numeric (e.g., `c(1, 2, 3)`), character (e.g., `c("word1", "word2")`), or boolean (e.g., `c(TRUE, FALSE, TRUE)`). In machine learning, vector representations, such as word embeddings, map text data into numerical spaces for analysis. Vectors can be visualized and analyzed by historians. For example, each point in a scatter plot can represent a vector. This allows researchers to identify patterns or trends in the data that may not be easily discernible from raw text. In text analysis, visualizing word embeddings as vectors can help uncover semantic relationships, such as words that are closer together in the plot being more contextually related.

Validation
An iterative process that involves assessing the accuracy, reliability, and relevance of claims derived from historical data or models. It requires evaluating the results of an analysis as well as returning to the original sources from which the data was drawn. This process often involves cross-referencing claims with primary sources, such as archival materials, speeches, or official documents, to check that the conclusions

align with the historical record. Historians often validate their findings by consulting secondary sources, such as scholarly articles and books, which provide context and alternative interpretations. This cyclical process of validation-—comparing computational results with traditional methods—-contexutally grounds claims about history, helping to safeguard against misinterpretation or the overgeneralization of data and historical events.

Visualization
The use of charts, graphs, maps, word-frequency plots, and other visual displays to make data easier to understand. Visualization helps researchers identify patterns, trends, and relationships that may be difficult to see in raw data alone. Visualization is both an exploratory tool for discovering patterns and a communication tool for presenting findings.

Vocabulary
The set of unique words or terms that characterize a corpus. They play an important role in guiding historical analysis. Historians can, for example, focus their analysis by examining the natural vocabulary embedded within texts or by curating their own set of terms to steer the results of algorithms. In computational historical analysis, historians often define a set of keywords or topics to track across a corpus, concentrating on terms that represent important historical concepts, such as "democracy," "empire," or "class." This vocabulary can be used to analyze trends over time, compare term usage across regions or political groups, or identify shifts in how key ideas are discussed.

Word Count (Raw)
A simple text metric that indicates the total number of words in a document or corpus, without adjustments for stop words, stemming, or lemmatization. This metric provides a basic measure of the length or size of a text and is often used as a foundation for more complex text analyses. While raw word count can offer useful perspectives, they do not account for semantic meaning, meaning that word counts can be less informative for deeper textual analysis. However, word counts are helpful in providing a preliminary overview of the text's volume, which can guide further analysis.

X-Axis, Y-Axis
The X-axis and Y-axis are the two perpendicular axes in a Cartesian coordinate system used for plotting data. The X-axis (horizontal) typically represents the independent variable, such as time, categories, or other variables that are controlled or manipulated. The Y-axis (vertical) represents the dependent variable, which is the outcome being measured, such as frequency, magnitude, or quantity. For example, in a scatter plot of top speakers, the X-axis might represent years, while the Y-axis represents word occurrences.

Terms Not Used in This Work

Text Mining for Historical Analysis follows a methodological framework that emphasizes careful interpretation grounded in primary and secondary sources. As a result, some terms are intentionally avoided because they are associated with research traditions that pursue different goals. Many of these terms originate from academic approaches that prioritize prediction, automation, or large-scale pattern detection rather than the context-sensitive, interpretive analysis that guides this book.

For example, "culturomics" and "big history" represent two distinct intellectual traditions that emphasize large-scale, often quantitative analyses of culture and history but do not prioritize the nuances of historical interpretation or the specific, localized examples that this work focuses on. While quantitative history applies statistical and computational techniques to historical data, **Text Mining for Historical Analysis** emphasizes the interpretive and critical dimensions of engaging historical data, such as by forming research processes guided by both close reading and quantiative modeling, rather than relying solely on statistical modeling to arrive at conclusions.

Here we define these terms:

Culturomics
A quantitative approach to studying cultural trends through large-scale textual analysis. Practitioners

of culturomics aim to uncover statistical patterns in language and culture, such as how certain words or ideas gain or lose prominence. This methodology focuses on identifying long-term trends and correlations in language use, often across broad societal shifts or major historical events. While this approach sounds similar to computational text analysis or corpus linguistics, it tends to emphasize quantitative measurements—such as frequency counts—without delving into the deeper, interpretive or contextual aspects of why these changes occur or what this change might mean for human history. It does not seek the qualitative insights that come from historians understanding the underlying social forces shaping language and history. As such, this book does not adopt culturomics' approach, as this book emphasizes a more context-based analysis of historical data situated in close reading rather than focusing solely on statistical patterns.

Big History
A multidisciplinary framework that examines history on an expansive scale and in a way that integrates insights from cosmology, geology, biology, and the social sciences to explore long-term patterns of change across the universe, Earth, and human history. Big History attempts to provide a grand narrative that spans billions of years, from the origins of the universe to the present day, by identifying broad, overarching trends that shape the course of history. While this book engages with historical patterns, it does not adopt the totalizing scope of Big History, which often emphasizes global, universal processes over specific historical contexts or localized events. Instead, this book focuses on more focused, context-sensitive historical analysis, emphasizing the complexities and specificity of a specific human history rather than attempting to unify all of history into one large-scale narrative. While quantitative readings inherently reduce the complexity of history into data points, we believe a narrative of this scale can be overly reductive.

Quantitative History
A methodological approach that uses statistical and computational techniques to analyze historical data, particularly large datasets. Often associated with economic and social history, it is commonly used to study patterns such as population change, economic development, and social movements. While *Text Mining for Historical Analysis* incorporates quantitative methods, its primary goal is historical interpretation rather than statistical modeling alone. Quantitative evidence can help identify patterns and trends, but understanding their significance requires attention to the social, cultural, and political contexts in which they emerged.

Cultural Evolution
A theoretical framework that applies evolutionary principles—-such as variation, selection, and inheritance—-to the process of cultural change. It draws parallels between biological evolution and the way cultures adapt and pass on ideas, practices, and technologies over time. While data science approaches can model cultural shifts and identify patterns in how culture changes, this book does not frame these changes in evolutionary terms. Framing cultural change strictly in evolutionary terms risks oversimplifying complex social and historical outcomes, reducing cultural shifts to a linear progression. It also overlooks the role of power dynamics, perhaps fostering the false impression that past cultures were inherently "simpler"—and that this supposed simplicity explains their disappearance—while implicitly portraying the present as more advanced or complex by comparison. Such an angle to thinking about history ignores the diverse range of human experiences that influence cultural change.

Metanarrative
A term from critical theory, particularly associated with Jean-François Lyotard, referring to overarching, totalizing explanations of history or knowledge that attempt to provide a single, unified narrative for all events or experiences. Such grand narratives often simplify complex histories by imposing broad, universal frameworks that can obscure the diversity of perspectives within historical nuance. This work instead proposes analytic paradigms through which the application of multiple algorithms and metrics allow for the analysis of historical data in a way that can reveal the affordances and limitations of an algorithm or model. By employing diverse computational techniques—-such as various text analysis algorithms and models-—this approach supports analysts exploration of different aspects of historical data, suggesting that analyzing trends and nuance go hand-in-hand, ensuring that multiple perspectives are considered in the analysis of historical events.

By clarifying these distinctions, we acknowledge the breadth of computational and data science approaches to the humanities and social sciences. These different approaches encompass a wide range of methodologies, theories, and intellectual traditions. While these diverse approaches each contribute valuable insights, we argue that it is important to situate computational historical analyses within a specific methodological framework that emphasizes critical analysis, contextual interpretation, and the nuanced understanding of historical data. By doing so, we seek to balance the abstraction approaches of data science with the richness of historical context so that digital methods serve as a tool for deeper understanding of history.